FloodRoute / ResilienceOS — Three-Person Build Plan

Hackachusetts 2026 · Team handoff · 24 October 2026
Status: Implementation plan, not a completed or validated geographic model.
Companion documents: `PROJECT_CONTEXT.md` (existing UI/data-contract brief) and `FloodRoute_3D_Feasibility_and_Sources.pdf` (source/risk audit). **This document refines the product scope**: it adds real ranking, intervention counterfactuals, and engineering investigation to the original semi-clickable prototype.

1. Product in one minute

Problem. Coastal flood maps tell Boston where water may reach, but do not directly tell planners which flooded roads most damage access to hospitals, fire stations, or communities; whether viable detours exist; or which roads deserve engineering investigation first.

Proposed solution. Analyze observed historical flood evidence alongside *separately labeled* future flood scenarios. Overlay potential road disruption on a routable street graph. Rank vulnerable **roads or corridors** by how much maintaining access would improve important journeys, including alternatives and demand. For a selected corridor, show the flood evidence and road characteristics; an evidence-grounded engineering copilot proposes options to *investigate*. Finally, allow a planner to **test an assumed protection**, rerun accessibility, and update all priorities. A proposed measure is not a verified physical fix.

Five-beat live demo:

1. Open an interactive **3D Boston** scene and choose a real historical-event footprint or published future flood scenario.

2. Show **normal vs flood-affected accessibility** for representative fire/hospital/community journeys.
3. Reveal the **highest-impact bottlenecks**, explaining why missing alternatives matter more than flood exposure alone.
4. Click a corridor to see evidence, road details, candidate measures, reference documents, conceptual cost context, and unknowns.
5. Choose **“assume this corridor remains passable”** and observe graph-based **reranking of the entire candidate set**. Explain that physical effectiveness has not been proven.

Not the product: a true real-time ambulance navigator; a live flood-physics engine; a validated road-level historical flow animation; an automated civil engineer; or a site-specific construction cost calculator.

Ownership at a glance

Owner	Workstream	Must deliver	Owns integration surface
Person 1	**Road graph, impact ranking, counterfactual reranking**	Connected graph, representative journeys, real computed comparison/ranking, re-ranking under assumed repair	`road_id`, `RoadRankingResult`, `CounterfactualResult`
Person 2	**Flood evidence, hazard mapping, 3D application and final UI integration**	At least one geographically grounded flood overlay, affected road mapping, responsive 3D UI and clickable story	`FloodExposureResult`, map scene, app composition
Person 3	**Road facts, engineering evidence, assistant, and investigation UX**	Source-backed corridor investigation with candidate options, caveated cost examples, intervention form	`InvestigationResult`, `InterventionProposal`

Coordination: Person 1 freezes canonical road IDs; Person 2 is the **frontend/integration owner**; Person 3 owns engineering documents and the investigation panel. None needs another person’s finished algorithm to begin: each starts from the agreed **fixture JSON** defined below.

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2. Common data contract — agree before parallel work

Shared identifiers and conventions

- `road_id`: stable within a **versioned canonical road manifest** (`graph_version`); never use an unrecorded CSV row number as a permanent ID. The starter Boston street CSV supplies WKT geometry and names but **no canonical unique street-segment ID**. Person 1 generates a manifest; everyone else uses those IDs. A future source remap must be explicit.
- `scenario_id`: for example `usgs-2018-jan-event` (historical evidence) versus `mcfrm-2050-1pct` (future annual-exceedance scenario). Never call a future scenario a past occurrence.
- `facility_id`, `journey_id`, `corridor_id` (one corridor may include multiple directed graph edges), `intervention_id`.
- UI geometry is **GeoJSON EPSG:4326 longitude/latitude**, not projected source geometry. Preserve each dataset's original coordinate reference system separately during analysis. Distances in meters, modeled travel time in seconds, elevations/depths with **explicit datum and units**.
- Every record carries `source_id`, `source_url` or `source_ref`, `data_status` (`observed`, `published_model`, `derived`, `mock`, `assumed`), and an explanation of limitations. Missing depth is `null`, never zero.

Minimal exchange contracts (names may be adjusted once, then frozen):

```
RoadManifest = { graph_version, roads: GeoJSONFeatureCollection, id_mapping, metadata }
FloodExposureResult = { scenario_id, graph_version, affected_roads: [
  { road_id, exposure: 'exposed'|'not_exposed'|'unknown',
    depth_m?: number|null, passability: 'potentially_closed'|'unknown'|'assumed_open',
    basis: 'official_extent'|'official_depth'|'historical_footprint'|'mock', source_id }
], extent_geojson, provenance }
RoadRankingResult = { scenario_set_id, graph_version, ranking: [
  { corridor_id, road_ids, priority_rank, impact_metric, metric_units,
    affected_journeys, detour_or_disconnect_evidence, assumptions, provenance }
] }
InvestigationResult = { corridor_id, road_ids, road_facts, flood_evidence,
  network_impact, candidate_methods, published_cost_examples,
  missing_site_information, citations, limitations }
InterventionProposal = { intervention_id, corridor_id, road_ids,
  proposed_method, status: 'hypothetical'|'implemented'|'verified',
  assumed_passability_by_scenario, review_notes, source_ids }
CounterfactualResult = { intervention_id, graph_version,
  before_ranking, after_ranking, accessibility_delta, assumptions }
```

Critical boundary: the ranking engine consumes **scenario-specific road exposure/passability assumptions**, not raw flood raster files or Cesium entities. The engineering assistant consumes **normalized facts and references**, not internal NetworkX graph objects. React never imports a backend script or raw CSV.

Repository proposal — file ownership is non-overlapping:

```
contracts/                  # Frozen shared JSON schema/TS types — Person 1 creates first
fixtures/                   # Small jointly agreed example, then read-only
analysis/network/           # Person 1: graph, trips, ranking, reranking, CLI/tests
analysis/hazard/            # Person 2: MC-FRM/USGS loaders, CRS, road exposure mapping
frontend/                   # Person 2: Cesium, main panels, composition, deployment
frontend/src/investigation/  # Person 3 only: investigation drawer and chat UI
engineering/                # Person 3: road facts, evidence index, cost precedents, prompts/tests
data/normalized/roads/      # Person 1: canonical road manifest and ID mapping
results/                    # Contract-compliant JSON outputs, owned by producing stream
```

One-minute handshake fixture: Before any external download, agree on **one area** (Morrissey Boulevard as candidate, subject to real flood overlap), **one scenario**, **two or three origin–destination pairs**, **one candidate corridor**, and a 20–50-segment fixture. This lets each stream implement and test independently. The fixture is illustrative and must say so.

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3. Person 1 — Road-network analysis, ranking, reranking

Mission

Produce a defensible **relative criticality ranking** for exposed roads/corridors based on **loss of access to important destinations and quality of alternatives**, then recompute priorities when an improvement is *assumed* or later verified. This person is accountable for the real quantitative core—not the LLM or 3D appearance.

Datasets and exact sources

1. **Starter:** `robotics.zip/boston-streets/data.csv` — **19,440** Boston street centerlines; `street`, `speed_mph`, `oneway`, `cost_forward`, `cost_back`, WKT `geometry`. **No stable segment ID**, no guaranteed topology; build a versioned graph and validate directionality. Source: Boston city SAM network via CKAN. This endpoint’s example request returns 100 rows; the ZIP has the larger saved snapshot.

2. **Fallback/richer network:** MassGIS–MassDOT Roads. Consider OSMnx/OpenStreetMap if graph topology proves too difficult; log source and version. Do **not** silently mix inconsistent road IDs.
3. **Destination facilities:** MassGIS Hospitals (August 2026) — point coordinates with `EMERGENCY_DEPT` flag and trauma attributes. The starter `healthcare.zip/hospitals/data.csv` has 103 rows but **no coordinates / ED flag**, so is insufficient by itself. MassGIS Fire Stations, Boston Public Safety GIS, fire departments layer 2.
4. **Service assignments (optional):** MassGIS Emergency Service Zones — jurisdiction/service areas, **not an ambulance movement feed**.
5. **Demand/context:** `civic.zip/neighborhoods/data.csv` — 24 rows of 2025 neighborhood estimates. **No geographic boundaries in this CSV.** To weight origin points, obtain matching neighborhood polygons or more granular Census data and document how origin demand was assigned.
6. **Scenario closures:** Person 2's `FloodExposureResult`. Work immediately with a mock closure fixture rather than waiting for hazard GIS processing.

Suggested implementation steps

- Parse streets/WKT; establish directed topology and weight interpretation (`cost_forward`, `cost_back` might already encode routing costs). Audit disconnected components, snap tolerances, bridge/tunnel behavior, and at least a few plausible local trips.
- Select **representative planning journeys** (e.g., a population-weighted neighborhood origin to an emergency department; fire station to a representative incident point). Avoid presenting arbitrary nearest-hospital journeys as clinically appropriate ambulance dispatch.
- Compute baseline accessibility; remove or heavily penalize **potentially impassable edges under explicit scenario assumptions**; recompute routes. Track detour time, distance, and **unreachable** separately. Unreachable must not be silently dropped from the score.

- Evaluate the marginal benefit of preserving **one road/corridor**, where feasible, conditional on all other scenario closures. Check alternate routes and serving multiple facilities automatically. Evaluate **corridor bundles**, since reopening one edge may be useless if adjacent edges remain flooded.
- Rank candidate **investigations**, not declare construction priorities or definitive fixes. Store which journeys and assumptions produced each ranking.
- Accept `InterventionProposal` from Person 3 (or fixture), update the assumed scenario-specific passability in a **sandbox copy** of the graph, and recompute *all* candidate rankings. A saved implemented/verified record requires actual evidence, not just a button click.

Candidate methods (intentionally provisional)

The first usable metric **could** be population/service-weighted **change in modeled travel time** from baseline to a flood scenario, with an explicit penalty or separate count for inaccessible journeys. For each candidate corridor, **consider** comparing total accessibility loss before and after hypothetically preserving that corridor across relevant scenarios. Scenario weights **might** reflect a curated set of cases or defensible annual probabilities; historical counts and future exceedance probabilities **must not** be combined as if they were the same thing.

A simple prototype can use Dijkstra or A* on a directed NetworkX graph. If runtime allows, test **redundancy** via second-best routes / alternate paths rather than manually giving every road an arbitrary "importance" score. These are proposals to validate, **not a locked formula**. There may be interaction effects: protecting A and B together can have a benefit that neither alone delivers.

Acceptance criteria

- Real graph builds from a documented input snapshot; ≥3 plausible selected trips produce inspectable baseline paths.
- Under one explicit mock or official flood-closure set, impacted journeys and a **computed**, explainable ranking change.

- A hypothetical intervention changes graph passability and recomputes rankings, **including effects on other corridors**.
- CLI or test harness produces valid `RoadRankingResult` and `CounterfactualResult` JSON using the shared IDs; runs without UI or engineering assistant.
- Unit tests cover an alternate-route example, an isolated neighborhood, and a multi-edge corridor case.

Defer: real traffic and signal timing; live dispatch; citywide origin sampling; calibrated monetary loss or mortality forecasts; sophisticated network portfolio optimization.

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4. Person 2 — Flood scenarios, geospatial mapping, 3D demo, integration

Mission

Ground the scenario in **actual flood geography** and create the central 3D interaction. Produce road exposure records for Person 1, **without implying exact road drivability or historical water-flow vectors**. Own the stitched user experience and final working local app.

Datasets and exact sources

1. **Published future hazard:** Massachusetts Sea Level Rise and Coastal Flooding Viewer / MC-FRM. Extents and depths for **2030, 2050, 2070** with specified annual exceedance probability; scenario probability is not a predicted storm date. MC-FRM technical overview. **Actual depth raster availability/download and licensing must be checked** before claiming precision.
2. **Observed historical event data:** USGS January & March 2018 inundation shapefiles, and USGS 2018 flood documentation. Footprints and high-water marks are **not a continuous high-resolution 3D flow recording**.
3. **Time series for contextual replay:** `climate.zip/tides/data.csv` — **744 hourly readings** in August 2024 at NOAA Boston station 8443970, **meters relative to MLLW**. To replay January 2018, fetch

the historical interval from NOAA CO-OPS API instead of substituting the ZIP's August 2024 values. Gauge water level is **not street-level water depth**.

4. **Terrain:** MassGIS LiDAR Terrain Data. Road-grade estimates from terrain must handle underpasses, bridges, datum, and survey limitations.
5. **Road geometry:** Person 1's versioned canonical manifest from `robotics.zip/boston-streets`. In parallel, begin 3D work using a small fixture in the same agreed shape.
6. **Renderer:** CesiumJS and streamed OSM/terrain assets when permitted. Cesium ion may need a token and internet; build graceful fallback if absent.

Suggested implementation steps

- Pick one **demonstrably affected** Boston corridor using public hazard/event geometry, not one selected only because it looks cinematic. Morrissey Boulevard is a plausible candidate, but verify overlap and resulting journeys first.
- Fetch one official flood **extent** and align CRS. Optionally add verified depth data; preserve flood datum metadata. Intersect it with the canonical street graph and return `FloodExposureResult`, including **unknown** passability when depth/road elevation is not enough.
- Build a Cesium scene with buildings, 3D terrain, scenario overlay, road colors, normal-vs-flood route polylines, camera fly-to and selected corridor detail.
- Implement a clearly labeled **visual replay**: for historic events, show observed footprint and measured water-level time series; only animate inferred spatial progression as **illustrative**, never as the exact historical fluid flow, direction, or velocity.
- Compose the ranking table from Person 1 and the engineering drawer from Person 3, initially using fixtures. Ensure clicking a row selects the same `corridor_id` in map and investigation panel.
- Own local run/build, browser smoke test, video/screen-record fallback, and one concise demo script.

Acceptance criteria

- Runs with `npm install && npm run dev` (or documented equivalent) on a teammate's laptop; map is navigable when Cesium assets load.
- At least one official flood footprint or extent renders in the **correct geographic location**. Mock overlays clearly say mock.
- Exports `FloodExposureResult` keyed to manifest road IDs, or an honest documented blocker; does not claim an extent intersection proves road closure.
- Selecting scenario, road/corridor, rank row, and intervention produces consistent panels; fallback mode works without an external token.
- Historical vs projected events are unmistakably distinguished and source metadata is visible.

Defer: GPU fluid dynamics, exact flow direction/time at road level, citywide seamless depth animation, live hazard detections, automatic passability guarantee.

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5. Person 3 — Evidence-grounded engineering recommendations

Mission

Answer **"what could a civil engineer reasonably investigate at this location, why, and what evidence is missing?"** The assistant supports preliminary investigation, not design certification or site-specific bidding.

Datasets and exact sources

1. **Physical road attributes:** Boston MassDOT Road Inventory GIS, layer 14 and MassGIS–MassDOT Roads — available width/surface/class/volume fields should be checked **per record**; do not promise every field is populated. Use Person 1's road IDs and spatial join to source attributes. `robotics.zip/boston-streets` already supplies road names, one-way, speed, costs and WKT.

2. **Ground/structure context:** MassGIS LiDAR terrain. `robotics.zip/bridges/data.csv` gives **2022** bridge condition summary **without coordinates**; use full FHWA/MassDOT inventory before attaching bridge condition to a particular site.
3. **Intervention methods/cost precedent:** Boston Climate Resilient Design Guidelines. They cover vegetated berms, raised seawalls, raised roadways and deployable barriers, including design considerations, operations/maintenance, and **2018 conceptual example costs**. The previously verified examples are roughly **$2.3m for a 2,000-ft sample berm** and **$13.6m for a 2,000-ft sample raised roadway**, not 2026 bids and not universal per-foot prices.
4. **Real local precedent:** MassDOT Morrissey Boulevard corridor study (June 2025 report), MassDOT 2025 report announcement, Boston coastal resilience implementation. Site plans must be represented accurately; do not claim proposals are approved or built without verification.
5. **Inputs from team:** Person 1's vulnerability record (detours, affected journeys, alternative connectivity) and Person 2's flood/event record (source, extent/depth where available, uncertainty). Start with one fabricated but labeled corridor fixture.

Suggested implementation steps

- Normalize the source information into a small **cited engineering evidence library**: measure, when potentially applicable, why it might fail, required data, cost-example scope/year, and URL/page if possible.
- Build the **Road Investigation** context object from authoritative road attributes + flood evidence + ranked-network consequences. Keep each field's provenance; omit unavailable facts rather than hallucinating material, water direction, drainage capacity or underground utilities.
- Design an assistant prompt (optionally backed by a live model) that **uses only retrieved/source-linked evidence for factual claims**, separates sourced observations from hypotheses, and always states missing engineering data. A deterministic reference-backed answer is a valid fallback if live LLM integration fails.

- Offer 2–4 candidate **methods to evaluate**, not one guaranteed fix: e.g. road elevation, continuous barrier, deployable measure, drainage/site works, alternate access. Discuss applicability and constraints, avoiding promises that a barrier only at the road protects the network.
- Display **published example cost context** with price year, dimensions, excluded costs, and caveats. Cost estimation for the specific selected road is a separate engineering task. The model may suggest a **preliminary estimating method** but must not invent defensible local numbers.
- Let a planner record an **InterventionProposal** (method, corridor, which scenarios it is *assumed* to withstand, and status). Hand this to Person 1 for **idealized passability counterfactual**; do not modify flood extents or automatically label a road verified/fixed.
- Present a short exportable **engineer review brief**: reason for priority, documented flood exposure, road facts, candidate measures, precedent links, cost caveats, and site investigations needed.

Acceptance criteria

- One selected corridor returns a useful `InvestigationResult` using actual named official documents, with accessible links and known/unknown separation.
- Missing road materials or depth become **“not available / requires site survey”**, not invented facts.
- Historical construction-price examples identify **2018**, sample dimensions, and non-transferability.
- Assistant or scripted fallback answers questions about the selected corridor, offers candidates, and produces a typed `InterventionProposal`.
- Investigation module runs against **fixture** ranking/flood contexts even if Persons 1–2 aren’t finished.

Defer: engineering stamping/certification, custom hydraulics, exact site cost bids, open-ended ungrounded agent browsing, claiming physical benefit without a validated model.

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6. Integration plan — minimize cross-team waiting

Start together (15–25 minutes): Agree on demo region, 1–2 scenario IDs, 2–3 journeys, one corridor fixture, normalized geometry and units. **Person 1** checks in `contracts/` and initial `road_manifest.geojson`; **Person 2** checks in a standalone 3D scene shell; **Person 3** prepares source-backed engineering fixtures. Freeze contract **v0.1** and record `graph_version`.

Milestone	Person 1	Person 2	Person 3	Integration test
Parallel start	Build network using synthetic road closures	Load Cesium scene + official extent independently	Curate guidelines + one corridor brief	All 3 render/read the same fixture IDs
First vertical slice	Computed baseline vs impacted trips	Selected flood footprint and road overlay	Investigation panel with evidence	Select corridor -> matching rank, map, brief
Counterfactual slice	Recompute rankings after assumed passability change	Animate before/ after network state	Emit typed hypothetical intervention	Changing selected measure updates affected rankings
Final check	Test disconnected journeys / reranking	Run app, screenshots, fallback & demo	Audit citations, costs, unknowns	No unlabeled mock numbers or claims of exact physics

Do not wait for a teammate:

- Ranking works against `fixtures/mock_flood_exposure.json` from minute one.
- Hazard mapping tests can intersect against the frozen `road_manifest.geojson` without any ranking output.
- Engineering research uses `fixtures/sample_corridor_context.json` without running Cesium/NetworkX.
- The frontend consumes source-agnostic adapters; it initially imports validated JSON results, not Python APIs.
- Once official data arrives, replace the fixture producer, not the consuming UI.

Single integrator: Person 2 owns app composition and last-mile polish; Person 1 owns data contract changes (must notify everyone); Person 3 owns investigation and source validation. No parallel edits to `App.tsx` or the same CSS/config files.

Suggested final demo scope: one real area + **one** published future scenario + optionally **one** historical event + **three** representative journeys + **one** reranking counterfactual + **one** sourced investigation. More geographic coverage is a stretch goal, not a prerequisite.

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7. Risks, validation, and what we must not overclaim

Risk	Severity	Mitigation / owner
Road source contains WKT but no stable ID; topological gaps	**High**	P1 creates versioned manifest and connectivity tests; consider OSMnx fallback
Published flood extent intersects a bridge/tunnel but does not show road drivability	**High**	P2 labels road status *potentially affected*; P1 does sensitivity runs on closures
Historical footprints do not show minute-by-minute street flow/ depth	**High**	P2 uses honest evidence view + illustrative animation only; do not claim observed velocity vectors
2030/2050/2070 modeled scenarios mistaken for historical flood frequency	**High**	Both scenarios have explicit `data_status`, horizon, annual probability / chosen weight
No meaningful emergency detour in selected corridor	**High**	P1 & P2 jointly test actual flood/ network overlap before locking demo geography
Near-neighbor edges all flood; repairing one road yields no benefit	**Medium**	P1 supports multi-edge **corridor bundles** and checks network interaction effects
Road properties (material/width/ drainage) absent or dated	**High**	P3 marks missing; joins official layers only when verified
LLM invents engineering recommendations or exact costs	**High**	P3 evidence library, links, structured unknowns, safe fallback; human-review framing
Simulated project incorrectly treated as physically effective	**High**	`hypothetical` state and scenario-scoped assumed passability; never alter hydrology without model
Large depth rasters, CRS and vertical datum mismatches	**High**	P2 begin with vector extents; record datum; optional small clipped raster later
Cesium assets/token/network unavailable during pitch	**Medium**	P2 offline graceful fallback + screen recording of golden path
Scoring ignores population, alternative routes, or unreachable journeys	**High**	P1 test small counterexamples; show component explanations and limitations

Definition of done (minimum defensible end-to-end product)

> Publicly sourced flood **extent** positioned correctly on real Boston map, **or** transparent fallback labeled mock.
>
> Road graph has consistent IDs and at least three validated representative journeys.

Rank list is generated by code from scenario-specific graph changes; explains alternate routes, important destinations, and unreachable cases.

A hypothetical intervention **changes the graph and produces a new computed ranking**.

Engineer sees actual cited design precedents, missing site facts and cost **examples**, not invented construction estimates.

Scenarios clearly distinguish historical observed footprints, future models, and visual illustration.

UI has a complete clickable route: scenario -> rank -> road detail -> investigation -> assumed intervention -> reranking.

Each stream can run and test independently; documented `README` explains real vs mocked elements.

Two questions to resolve early, not silently assume

1. Does the chosen official flood extent overlap enough routable links to produce a **meaningful accessibility change**? If not, pick a different neighborhood; don't stage a false real-world result.
2. Can the team acquire useful **historical event footprints** for the chosen corridor? If not, focus the actual analysis on MC-FRM scenarios and show 2018 only as regional context.

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8. Source directory — verified documentation, integration not guaranteed

The following public pages/services were checked for **existence and described content** on October 24, 2026. This is **not** proof that every service endpoint has been downloaded, decoded or joined successfully.

ID	Authority / source	Purpose	URL
S01	Starter Robotics / City of Boston SAM	WKT roads, direction/ speed/cost	Boston data API
S02	MassGIS–MassDOT Roads	Full public roadway network	MassGIS roads
S03	MassGIS Hospitals, updated Aug 2026	Hospital coordinates, ED and trauma attributes	MassGIS hospitals
S04	MassGIS Fire Stations / Boston Public Safety	Fire response origins	MassGIS fire stations · Boston GIS layer
S05	MassGIS Emergency Service Zones	Service agency assignment geography	MassGIS ESZ

ID	Authority / source	Purpose	URL
S06	Massachusetts MC-FRM	Published future flood extents and depths	Flooding viewer · technical overview
S07	USGS coastal 2018 flooding	Historical footprints and high-water evidence	USGS map data · study
S08	NOAA Boston tide gauge (8443970)	Water levels at station, time context	CO-OPS API
S09	MassGIS LiDAR terrain	Bare-earth elevations, slope context	LiDAR
S10	City of Boston road inventory GIS	Road characteristics; verify completeness	Boston road inventory
S11	Boston climate-resilient design guidelines	Engineering options and 2018 sample costs	Design guidelines
S12	MassDOT Morrissey Boulevard study	Local flood/transport constraints and proposals	Final report
S13	City of Boston	Coastal resilience implementation status	Project tracker
S14	Census & starter Civic ZIP	Representative population/demand	Census portal

Final interpretation: The team can build a compelling **network-aware resilience investigation tool**, not an autonomous flood-engineering system. The strongest defensible output is a transparent priority queue and engineering review brief that updates when a hypothesized intervention changes connectivity.